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Abbreviations

NTU	Nanyang Technological University	MM	ACM International Conference on Multimedia
THU	Tsinghua University	ICCV	IEEE International Conference on Computer Vision
USTC	University of Science and Technology of China	AAAI	AAAI Conference on Artificial Intelligence
CVPR	Conference on Computer Vision and Pattern Recognition	TPAMI	IEEE Transactions on Pattern Analysis and Machine Intelligence
NeurIPS	Conference on Neural Information Processing Systems	TIP	IEEE Transactions on Image Processing
ICLR	International Conference on Learning Representations	IJCV	International Journal of Computer Vision
ACL	Annual Meeting of the Association for Computational Linguistics	TSG	IEEE Transactions on Smart Grid
		EE	Electrical Engineering

Research Interests

Foundation models, with interests in multimodal reasoning, agentic and affective intelligence, controllable and efficient diffusion generation, and reliable adaptation.
Robust learning and principled optimization, providing theoretical foundations for improving foundation models.

Working Experience

Professor, USTC, China — Mar 2026 – Present
Research Scientist, NTU, Singapore — Mar 2025 – Feb 2026
Researcher, SenseTime, Beijing — Jul 2019 – Nov 2020

Education

Ph.D. in Computer Science, NTU, Singapore — Jan 2021 – Jan 2025
supervisor: Prof. Hanwang Zhang
M.S. in EE, THU, Beijing. — Sep 2016 – Jul 2019
B.E. in EE, THU, Beijing. — Sep 2012 – Jul 2016

Selected Publications

Listed in chronological order. *: equal contribution. †: corresponding author.

First-Authored Publications

IJCV 2026
1. Generalized logit adjustment: Improved fine-tuning by mitigating label bias in zero-shot vision models Imbalanced Learning.
B. Zhu, *Q. Sun, X. Yang and H. Zhang*

TPAMI 2026
2. Parallel diffusion solver via residual dirichlet policy optimization.
R. Wang, Z. Li*,* ***B. Zhu****, L. Yuan, H. Zhang, X. Yang, X. Chang and C. Zhang*

ICLR 2026
3. Reducing class-wise performance disparity via margin regularization.
B. Zhu, *K. Zhao, J. Cui, Q. Sun, Y. Zhou, X. Yang and H. Zhang*

NeurIPS 2025
4. Adaptive stochastic coefficients for accelerating diffusion sampling.
R. Wang,* ***B. Zhu****, J. Li, L. Yuan and C. Zhang*

ICCV 2025
5. Distilling parallel gradients for fast ODE solvers of diffusion models.
B. Zhu*, R. Wang*, T. Zhao, H. Zhang and C. Zhang*

CVPR 2025 **Highlight (3.0%)**
6. Project-probe-aggregate: efficient fine-tuning for group robustness.
B. Zhu, *J. Cui, H. Zhang and C. Zhang*

NeurIPS 2024
7. Robust fine-tuning of zero-shot models via variance reduction.
B. Zhu, *J. Cui and H. Zhang*

NeurIPS 2024 **Spotlight (2.1%)**
8. Enhancing zero-shot vision models by label-free prompt distribution learning and bias correcting.
X. Zhu,* ***B. Zhu****, Y. Tan, S. Wang, Y. Hao and H. Zhang*

MM 2024 **Oral (4.0%)**
9. Selective vision-Language subspace projection for few-shot CLIP.
X. Zhu,* ***B. Zhu****, Y. Tan, S. Wang, Y. Hao and H. Zhang*

NeurIPS 2023
10. Generalized logit adjustment: Calibrating fine-tuned models by removing label bias in foundation models.
B. Zhu, *K. Tang, Q. Sun and H. Zhang*

ICCV 2023
11. Prompt-aligned gradient for prompt tuning.
B. Zhu, *Y. Niu, Y. Han, Y. Wu and H. Zhang*

AAAI 2023 **Oral (10.8%)**
12. Debiased fine-tuning for vision-language models by prompt regularization.
B. Zhu, *Y. Niu, S. Lee, M. Hur and H. Zhang*

AAAI 2022 **Oral (4.2%)**
13. Cross-domain empirical risk minimization for unbiased long-tailed classification.
B. Zhu, *Y. Niu, X. Hua and H. Zhang*

TIP 2021
14. Structure-coherent deep feature learning for robust face alignment.

C. Lin, **B. Zhu***, Q. Wang, R. Liao, C. Qian, J. Lu and J. Zhou*

Corresponding-Authored Publications

NeurIPS 2026 **Oral (0.4%)**
15. Thinking with images as continuous actions: Numerical visual chain-of-thought.
*K. Zhao, †**B. Zhu**, J. Zhou, X. Zhu, Z. Yue and H. Zhang*

MM 2026 **Oral**
16. Layer roles matter: Mitigating modality imbalance in multimodal emotion reasoning via adaptive layer profiling.
*P. Huang, P. Song, †**B. Zhu**, Z. Han, J. Hu and X. Yang*

ICML 2026
17. Omni-perception policy optimization for multimodal emotion reasoning.
*Z. Han, †**B. Zhu**, W. Tong, P. Shao, P. Song, X. Wang, J. Chen, L. Lu and X. Yang*

CVPR 2026
18. Adapting point cloud analysis via multimodal Bayesian distribution learning.
*X. Zhu, L. Yi, S. Wang, W. Zhu, Y. Wu, †**B. Zhu** and H. Zhang*

TIP 2026
19. Hybrid granularity distribution estimation for few-shot learning: statistics transfer from categories and instances.
*S. Wang, T. Qi, X. Zhu, Y. Hao, †**B. Zhu**, H. Zhang and M. Wang*

ICLR 2026
20. Real-time motion-controllable autoregressive video diffusion.
*K. Zhao, J. Shi, †**B. Zhu**, J. Zhou, X. Shen, Y. Zhou, Q. Sun and H. Zhang*

ICLR 2026
21. Look carefully: adaptive visual reinforcements in multimodal large language models for hallucination mitigation.
*X. Zhu, K. Zhao, L. Yi, S. Wang, Z. Wang, †**B. Zhu**, H. Zhang and X. He*

AAAI 2026
22. DEPO: dual-efficiency preference optimization for LLM agents.
*S. Chen, M. Zhao, L. Xu, Y. Zhao, †**B. Zhu**, H. Zhang, S. Zhao and C. Lu*

MM 2025 **Oral**
23. Benchmarking and bridging emotion conflicts for multimodal emotion reasoning.
*Z. Han, †**B. Zhu**, Y. Xu, P. Song and X. Yang*

ICCV 2025
24. Unsupervised visual chain-of-thought reasoning via preference optimization.
*K. Zhao, †**B. Zhu**, Q. Sun and H. Zhang*

TSG 2019
25. Fault location for radial distribution network via topology and reclosure-generating traveling waves.
*S. Shi, †**B. Zhu**, A. Lei and X. Dong*

TSG 2018
26. Fault classification for transmission lines based on group sparse representation.
*S. Shi, †**B. Zhu**, S. Mirsaeidi and X. Dong*

Others

NeurIPS 2026
27. Beyond SFT-to-RL: Pre-alignment via black-box on-policy distillation for multimodal RL.
*S. Wang, W. Huang, X. Yu, Z. Yang, H. Lin, K. Wu, C. Xiao, W. Wang, C. Chen, **B. Zhu**, Y. Zhang and C. Qin*

NeurIPS 2026
28. TopoRefine: Topology-aware correspondence and residual refinement for training-free subject-consistent generation.
*Z. Gao, Z. Ti, C. Zhao, **B. Zhu**, X. Hu, Y. Ge, J. Yang and Y. Tai*

TPAMI 2026
29. Generalized kullback-leibler divergence loss.
*J. Cui, **B. Zhu**, Q. Xu, Z. Tian, X. Qi, B. Yu, H. Zhang and R. Hong*

TIP 2026
30. Detail++: training-free detail enhancer for text-to-image diffusion models
*L. Chen, J. Wang, Z. Pan, **B. Zhu**, X. Yang and C. Zhang*

ACL 2026 Findings
31. AMA: adaptive memory via multi-agent collaboration.
*W. Huang, Z. Wang, H. Lin, S. Wang, B. Xu, Q. Li, **B. Zhu**, L. Yang and C. Qin*

ACL 2026 **Oral (5%)**
32. Mitigating hallucinations in large vision-language models without performance degradation.
*X. Zhu, J. Fang, S. Wang, **B. Zhu**, Z. Wang, Y. Yang and X. He*

ICML 2026
33. Robustifying vision-language models via test-time prompt adaptation.
*X. Zhu, H. Wu, S. Wang, **B. Zhu**, J. Ge, J. Zhang and L. Chen*

CVPR 2026
34. Principled steering via null-space projection for jailbreak defense in vision-language models
*X. Zhu, **B. Zhu**, S. Wang, J. Fang, K. Zhao, H. Zhang and X. He*

CVPR 2026
35. Few-step diffusion sampling through instance-aware discretizations
*L. Yuan, R. Wang, T. Zhao, D. Fu, M. Lei, **B. Zhu** and C. Zhang*

CVPR 2026
36. ACE-merging: data-free model merging with adaptive covariance estimation
*B. Xu, H. Wu, H. Lin, W. Huang, **B. Zhu**, Y. Shu and C. Qin*

ICLR 2026
37. CoDi: subject-consistent and pose-diverse text-to-image generation.
*Z. Gao, **B. Zhu**, L. Yao, J, Yang and Y. Tai*

ICLR 2026
38. GuardAlign: robust safety alignment in multimodal large language models.

*X. Zhu, **B. Zhu**, J. Fang, S. Wang, Y. Zhang, X. Wang and X. He*

ICLR 2026 — 39. Free lunch for stabilizing rectified flow inversion.
*C. Wang, **B. Zhu** and C. Zhang*

ICLR 2026 — 40. Streaming drag-oriented interactive video manipulation: drag anything, anytime!
*J. Zhou, Y. Zhou, K. Zhao, Q. Xu, **B. Zhu**, R. Hong and H. Zhang*

AAAI 2026 — 41. Hierarchical semantic alignment for image clustering.
*X. Zhu, **B. Zhu**, Y. Li, J. Fang, S. Wang, K. Zhao and H. Zhang*

Preprint 2025 — 42. Generative distribution distillation.
*J. Cui, **B. Zhu**, Q. Xu, X. Xu, P. Chen, X. Qi, B. Yu, H. Zhang and R. Hong*

NeurIPS 2025 **Spotlight (3.2%)** — 43. Enhancing CLIP robustness via cross-modality alignment.
*X. Zhu, **B. Zhu**, S. Wang, K. Zhao and H. Zhang*

ICCV 2025 — 44. Dynamic multimodal prototype learning in vision-language models.
*X. Zhu, S. Wang, **B. Zhu**, M. Li, Y. Li, J. Fang, Z. Wang, D. Wang and H. Zhang*

CVPR 2025 — 45. Devils in middle layers of large vision-language models: interpreting, detecting and mitigating object hallucinations via attention lens.
*Z. Jiang, J. Chen, **B. Zhu**, T. Luo, Y. Shen and X. Yang*

CVPR 2025 — 46. Stylestudio: text-driven style transfer with selective control of style elements.
*M. Lei, X. Song, **B. Zhu**, H. Wang and C. Zhang*

CVPR 2024 — 47. Classes are not equal: an empirical study on image recognition fairness.
*J. Cui, **B. Zhu**, X. Wen, X. Qi, B. Yu and H. Zhang*

AAAI 2023 **Oral (10.8%)** — 48. Leveraging modality-specific representations for audio-visual speech recognition via reinforcement learning.
*C. Chen, Y. Hu, Q. Zhang, H. Zou, **B. Zhu** and E. Chng*

Patents

CN Patent CN122433924A — 1. 基于快慢思考协同的多模态情感推理方法和系统
杨勋；韩知渊；张嘉宇；**朱贝尔**；宋培培

CN Patent CN122414419A — 2. 基于强化学习与全模态感知优化的情感推理方法及系统
杨勋；韩知渊；张嘉宇；**朱贝尔**；宋培培

US Patent US11308710B2 — 3. Polygonal Region Detection.
***Zhu Beier** and Zhang Rui*

CN Patent CN110136156A — 4. 一种多边形区域检测方法及装置
朱贝尔；张睿

Service

Journal Reviewer: TPAMI, TIP, TMM, TCSVT, TCE, IJCV, Information Fusion

Conference Reviewer: CVPR, ICCV, AAAI, MM, UAI, AISTATS, NeurIPS (**Top Reviewer** 2025), ICLR, ICML (**Top Reviewer** 2026), ACL, KDD.

Awards

Best Paper, International Conference on AI for Engineering, China.	2026
Honorable Mention, Nanyang Speech Forum, Singapore.	2023
AISG PhD Fellowship, National Research Foundation, Singapore.	2021
First Price Scholarship, THU, China.	2018
Scholarship, China Scholarship Council, China.	2014

Internship

Research Intern, SenseTime, Beijing	Mar 2018 – Oct 2018
Research Intern, Tencent YouTu Lab, Shanghai	Jul 2018 – Sep 2018
Research Intern, MeiTuan, Beijing	Aug 2017 – Dec 2017

[*CV compiled on 2026-10-28*]